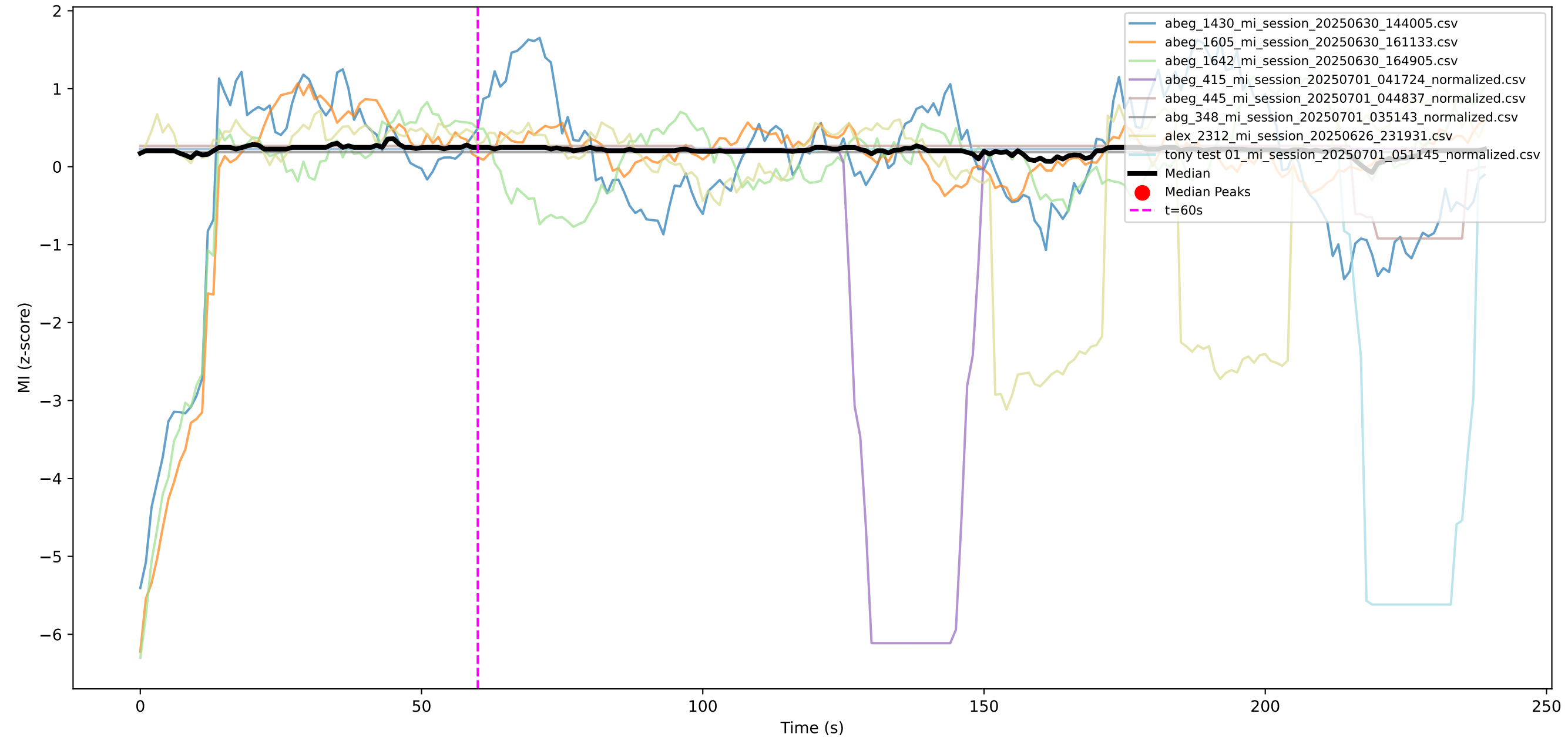
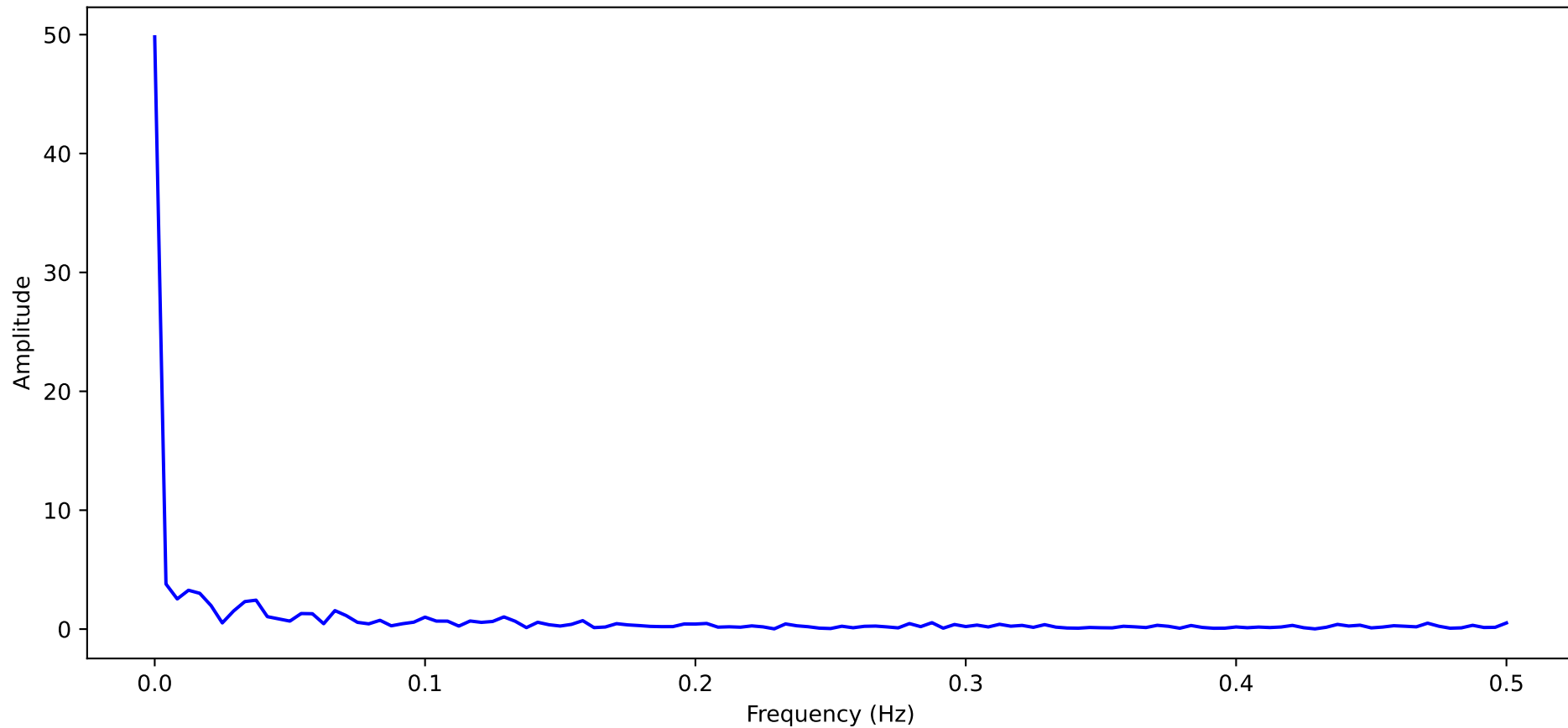


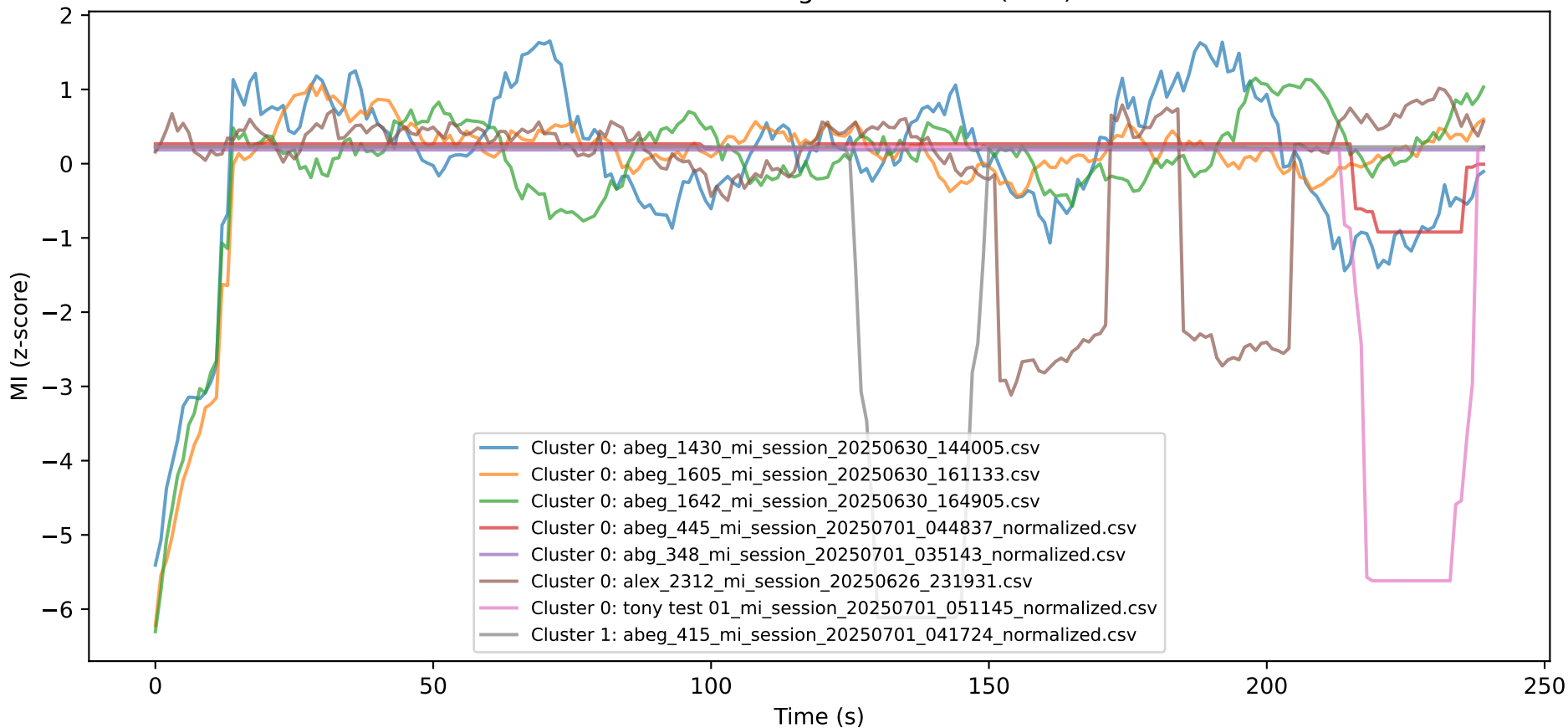
MI Curves (z-score, MA=20) with Median and Peaks



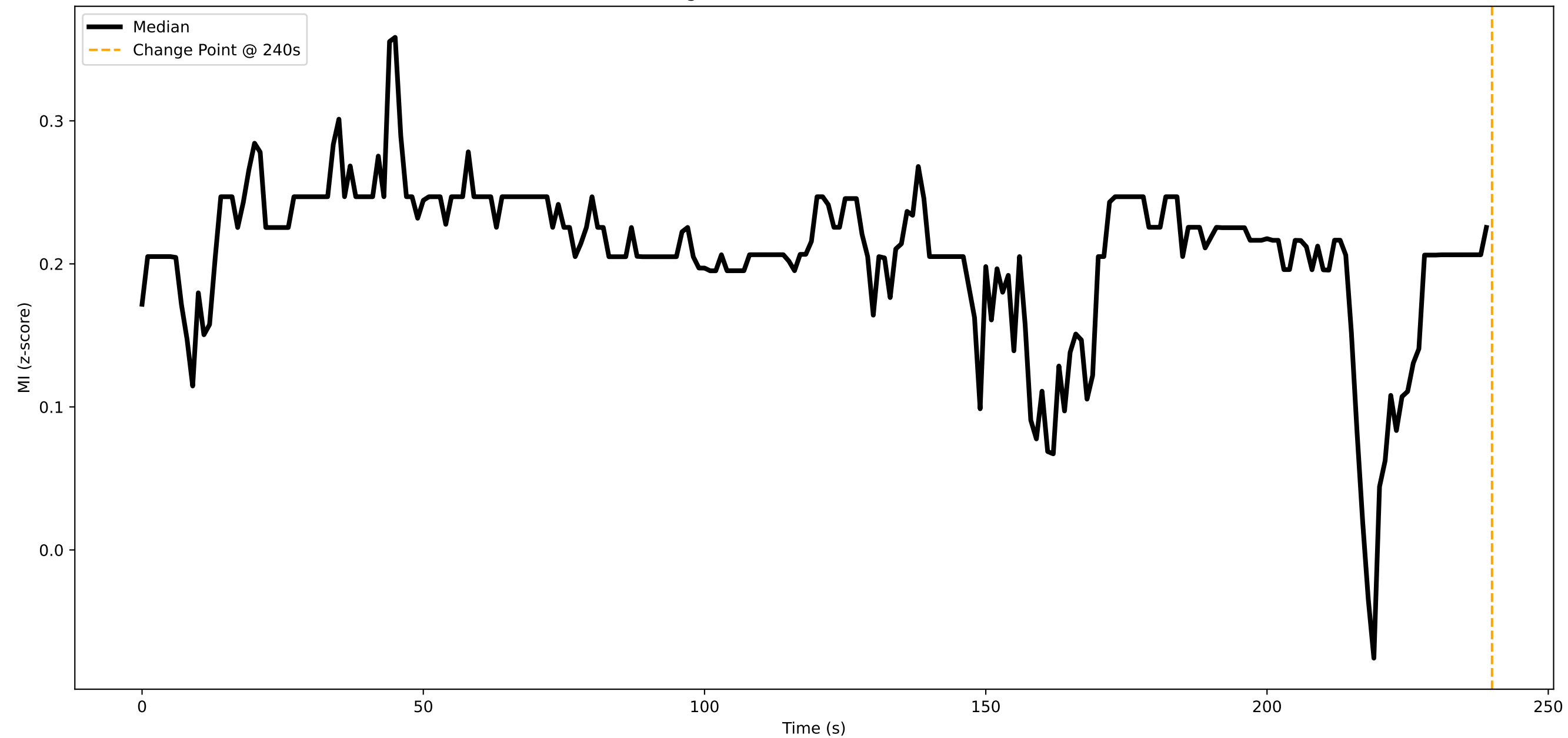
Median MI Frequency Spectrum (FFT)



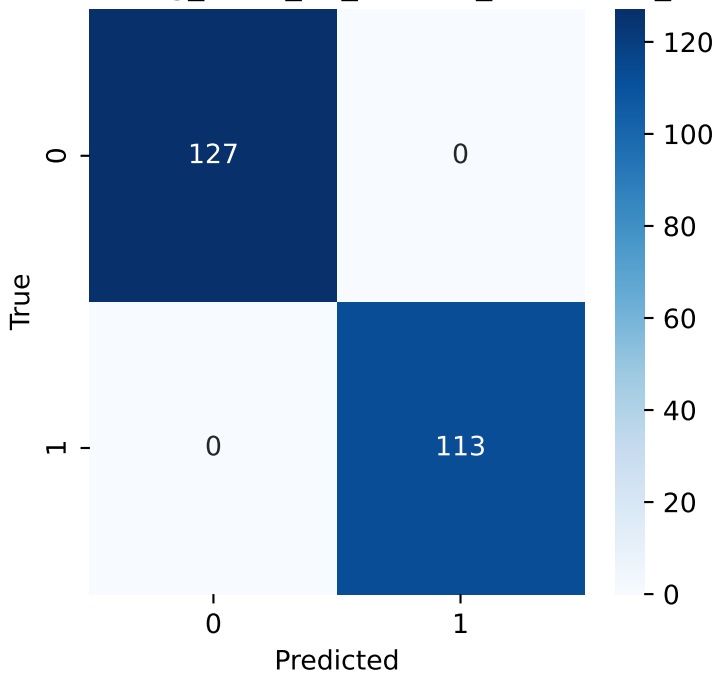
KMeans Clustering of MI Curves (k=2)



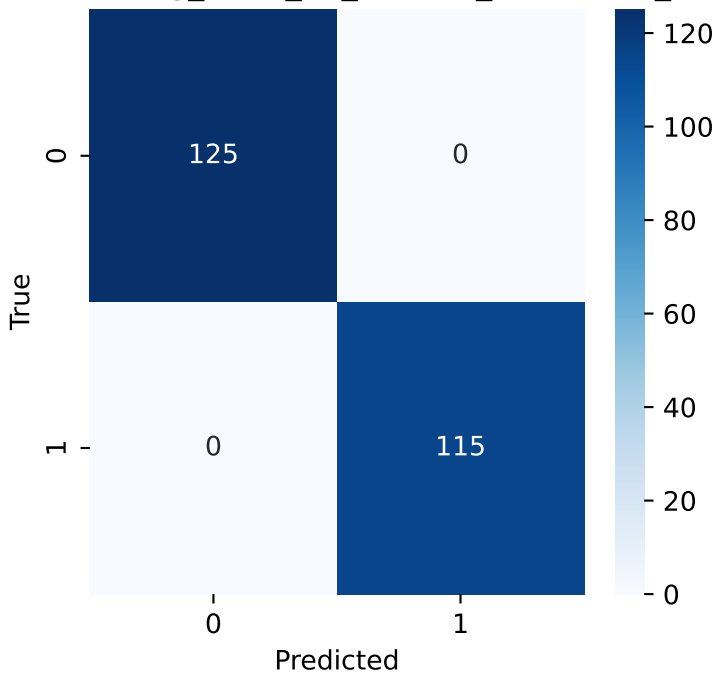
Change Point Detection on Median MI



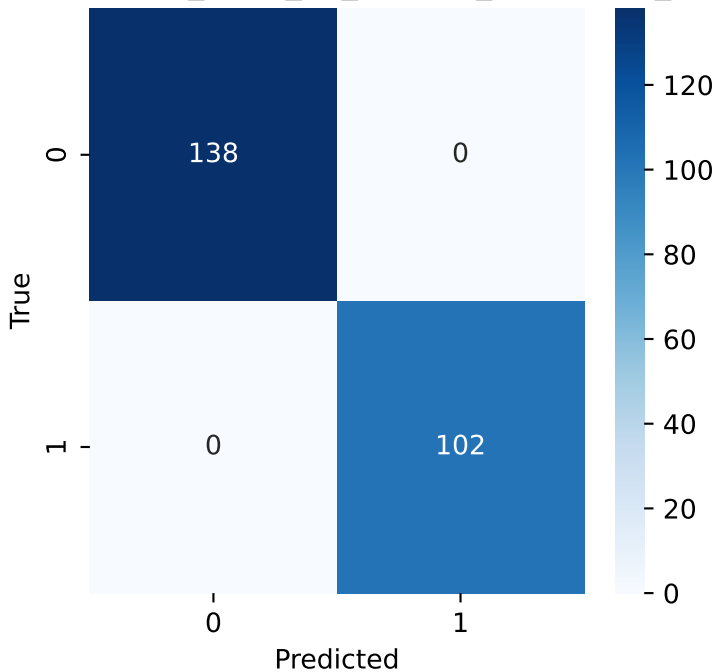
Matrix: abeg_1430_mi_session_20250630_1440



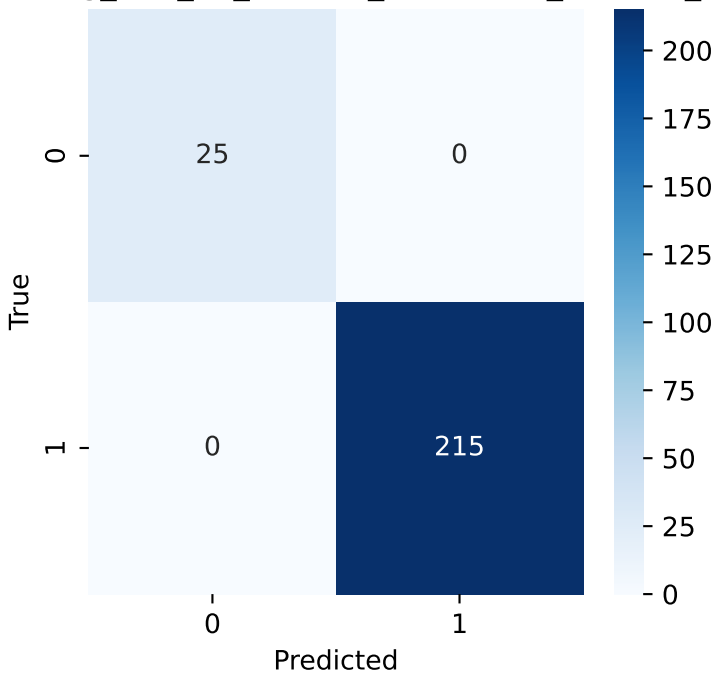
Matrix: abeg_1605_mi_session_20250630_1611



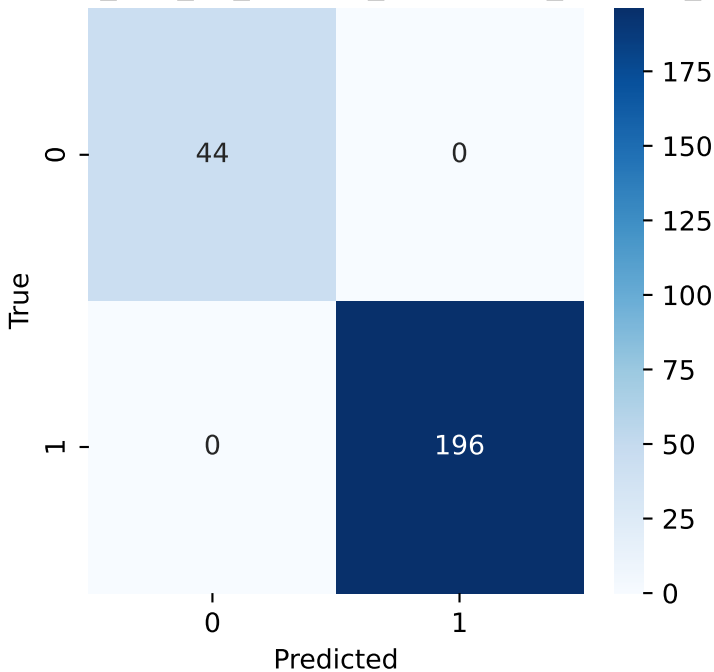
Matrix: abeg_1642_mi_session_20250630_1649



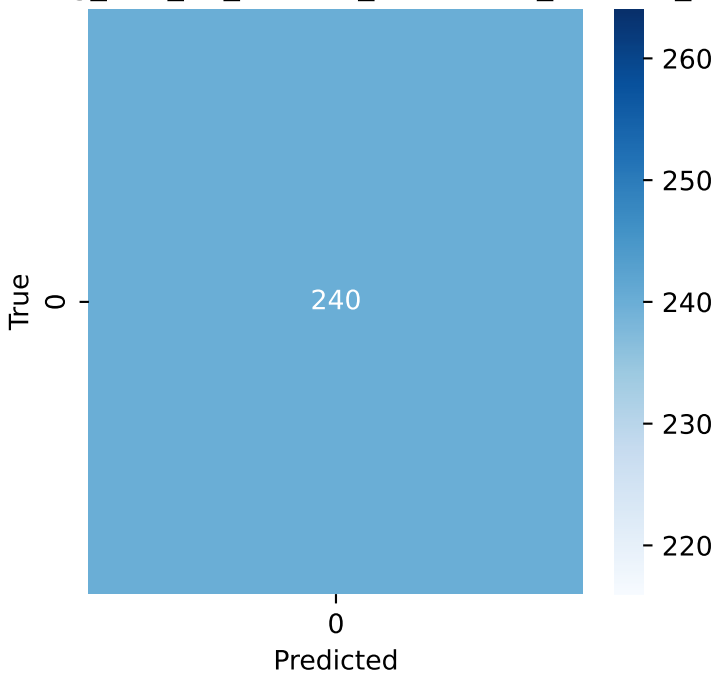
x: abeg_415_mi_session_20250701_041724_no



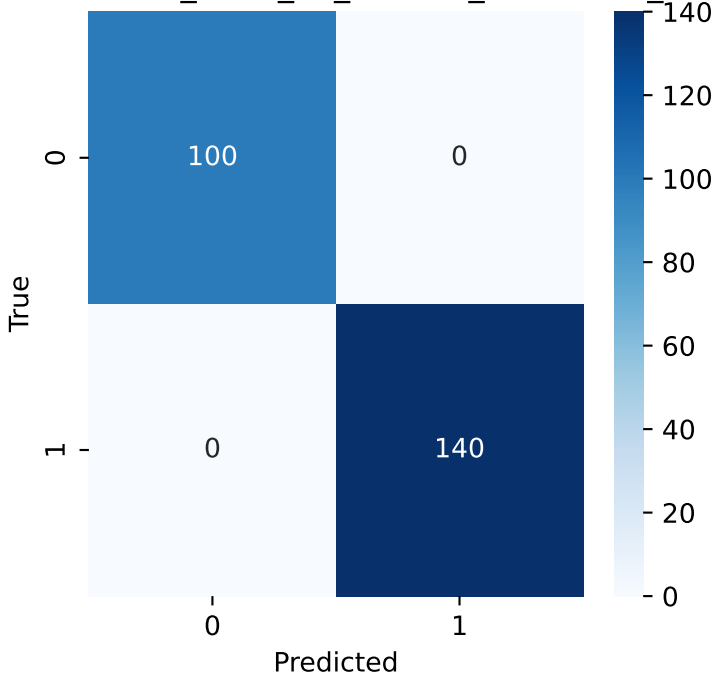
x: abeg_445_mi_session_20250701_044837_no



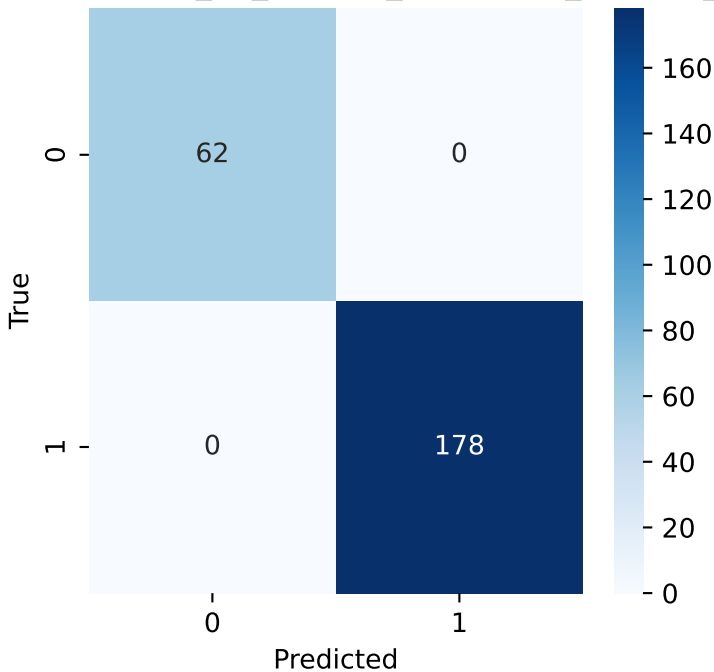
ix: abg_348_mi_session_20250701_035143_nor



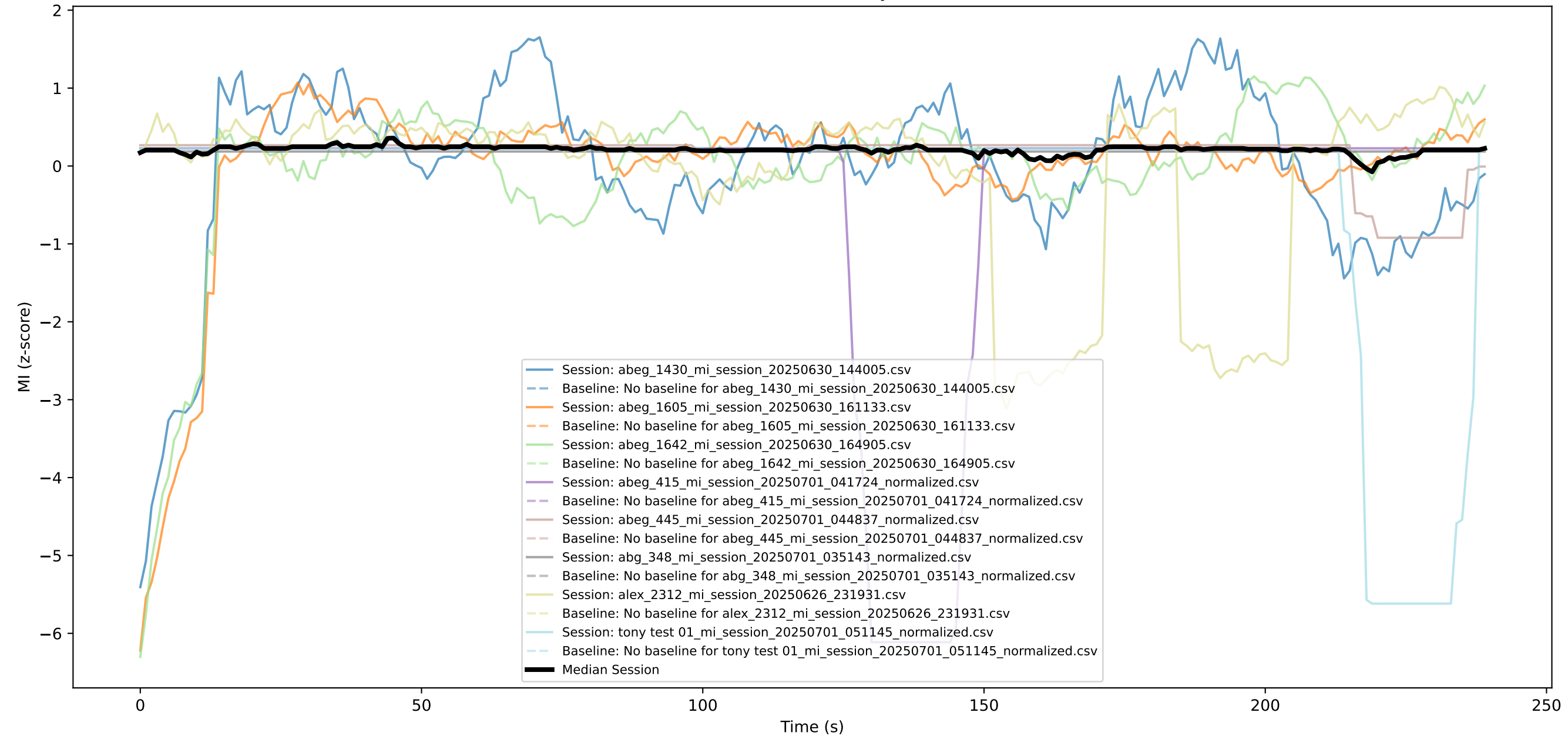
Matrix: alex_2312_mi_session_20250626_2319



ky: tony test 01_mi_session_20250701_051145_no



Session MI vs. Baseline (Eyes Closed)



MI Advanced Report
Generated: 20250717_002817

MI Advanced Report - Statistical Summary

1. Mindfulness Index (MI) Trajectory

- All MI session files with ≥ 240 samples were included.
- Individual and median MI curves were plotted (see Figure 8.1).
- RM-ANOVA across five epochs (0-30s, 31-90s, 91-150s, 151-240s, 241-300s):

Anova

	F Value	Num DF	Den DF	Pr > F
Epoch	nan	4.0000	28.0000	nan

- Post-hoc pairwise comparisons (Tukey HSD):
Multiple Comparison of Means - Tukey HSD, FWER=0.05

group1	group2	meandiff	p-adj	lower	upper	reject
0-30s	151-240s	0.2061	nan	nan	nan	False
0-30s	241-300s	nan	nan	nan	nan	False
0-30s	31-90s	0.6349	nan	nan	nan	False
0-30s	91-150s	0.2753	nan	nan	nan	False
151-240s	241-300s	nan	nan	nan	nan	False
151-240s	31-90s	0.4287	nan	nan	nan	False
151-240s	91-150s	0.0692	nan	nan	nan	False
241-300s	31-90s	nan	nan	nan	nan	False
241-300s	91-150s	nan	nan	nan	nan	False
31-90s	91-150s	-0.3596	nan	nan	nan	False

- Time-frequency analysis (FFT) of the median MI curve was performed (see Figure 8.3).

2. MI Dynamics

- Peak analysis (median curve): 0 peaks ($z > 1.5$) at times [], amplitudes [], durations [] (see Table 8.2, Fig. 8.2).
- AUC (median): 49.61; mean $\pm$ SD (individual): -10.53 ± 40.79 (see Table 8.3).
- Bootstrap 95% CI for mean MI difference (0-150s vs 151-300s): mean diff = 0.000, CI = [-0.014, 0.014].
- Change point detection (Pelt algorithm): 1 change points at [240] (see Figure 8.5).

3. Clustering & Mixed-Effects

- K-Means clustering (k=2): cluster averages {0: -0.011260313355158206, 1: -0.30154360553767134}, u
- Mixed-effects model (MI ~ time + (1|Subject)): Mixed Linear Model Regression Results

Model: MixedLM Dependent Variable: value

No. Observations:	1920	Method:	REML
No. Groups:	8	Scale:	1.2493
Min. group size:	240	Log-Likelihood:	-2953.1571
Max. group size:	240	Converged:	Yes
Mean group size:	240.0		

	Coef.	Std.Err.	z	P> z	[0.025	0.975]
Intercept	0.077	0.077	0.997	0.319	-0.075	0.229
time	-0.001	0.000	-2.835	0.005	-0.002	-0.000
Group Var	0.027	0.016				

=====

- ANOVA on final epoch by cluster: F=nan, p=nan (see Table 8.5).